

Table of Contents

1. Executive Summary		Page 3
2. Dataset Overview		Page 3
3. Exploratory Data Analysis (EDA)		Page 4
4. Customer Segmentation		Page 7
5. Demand Forecasting		Page 9
6. Churn Prediction		Page 12
7. Inventory Optimisation		Page 13
8. Interactive Dashboard & FastAPI REST API		Page 15
9. MLOps & Infrastructure		Page 16
10. Performance Summary & Conclusion		Page 16
11. Appendix: Complete Technology Stack		Page 17

1. Executive Summary

RetailPulse is a production-grade data science platform that transforms raw retail transaction data into actionable business intelligence. Using the **Online Retail II** dataset (UCI Machine Learning Repository, 1M+ transactions across 41 countries), the platform delivers four core machine learning capabilities through an interactive dashboard and REST API, supporting data-driven decisions to increase profitability and reduce operational waste.

Feature	Technology Stack	Target Criteria
Demand Forecasting	Prophet + LSTM Ensemble	MAPE $\leq$ 12% on 30-day horizon
Customer Segmentation	RFM + K-Means / DBSCAN	Silhouette Score $\geq$ 0.4
Churn Prediction	XGBoost + SHAP Interpretability	AUC-ROC $\geq$ 0.88
Inventory Optimisation	Safety Stock + Monte Carlo	Stockout Risk reduced by 30–50%

The platform includes a 9-page Streamlit dashboard with global filters, role-based access control, a FastAPI REST API with JWT authentication, MLflow experiment tracking, Evidently AI drift detection, Prefect orchestration, and full Docker/Kubernetes deployment support with Prometheus/Grafana monitoring.

2. Dataset Overview

The Online Retail II dataset contains real transaction history from a UK-based online retailer between December 2009 and December 2011. The raw dataset was processed and loaded into a structured schema consisting of the following core tables:

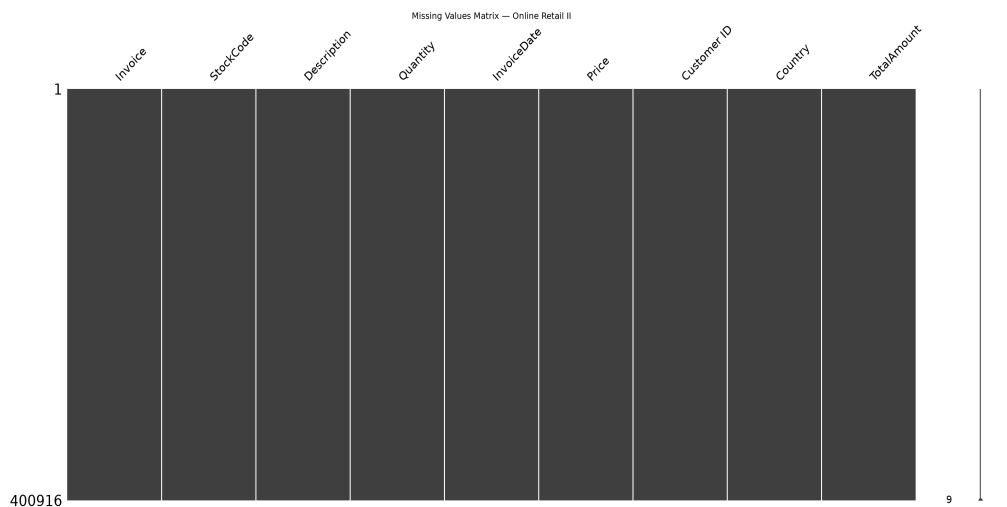
Attribute	Value / Target Details
Source	UCI Machine Learning Repository
Format	Excel (.xlsx) — 2 sheets representing 2009-2010 and 2010-2011
Total Rows	1,067,371 raw transaction rows
Unique Customers	5,931 distinct customer records
Date Range	December 2009 — December 2011
License	CC0 (Public Domain) — Permissive open license

3. Exploratory Data Analysis (EDA)

The data pipeline automatically cleans the raw dataset: removing transaction cancellations (invoices starting with 'C'), filtering out non-positive quantities and prices, dropping rows without a valid CustomerID, and resolving duplicate entries.

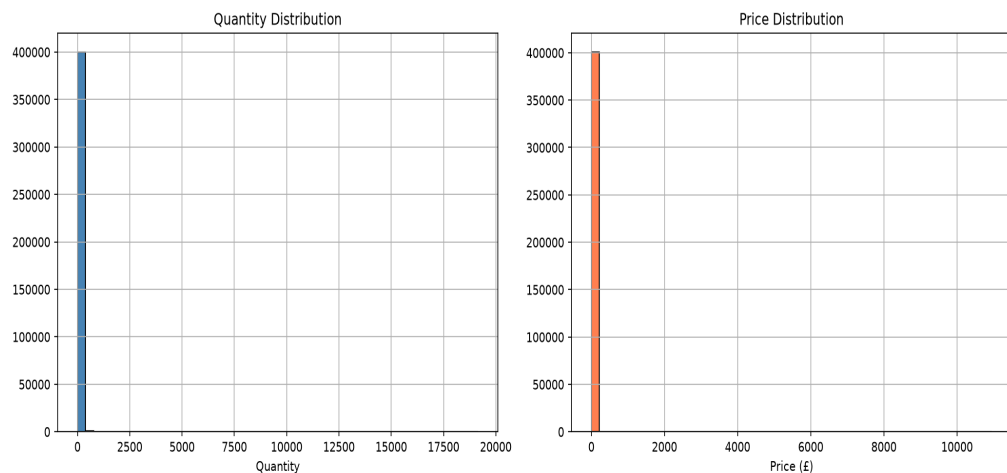
Data Quality & Null Value Distribution

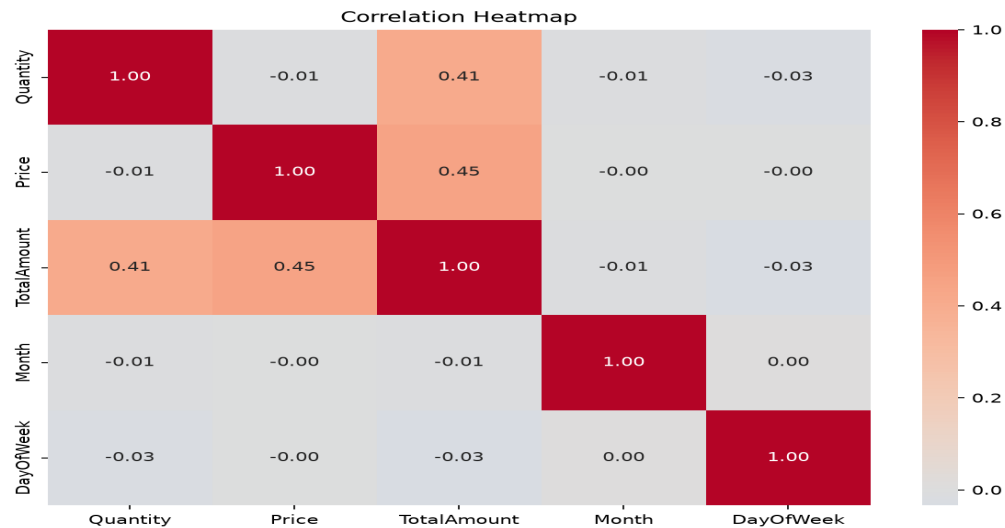
A data quality analysis indicates that CustomerID is the only column with significant missing entries (~25%). Because customer identification is critical for RFM segmentation and churn prediction, records with missing CustomerID are removed.



Distribution & Feature Correlations

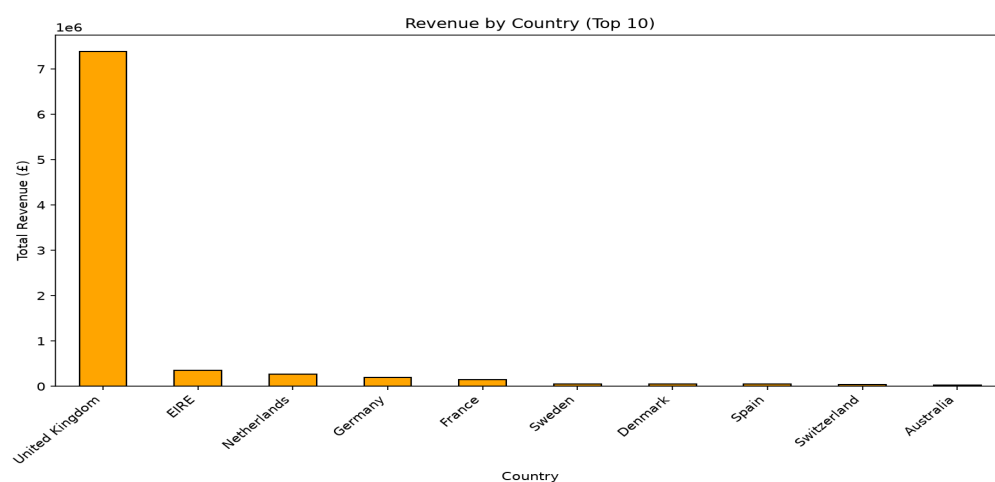
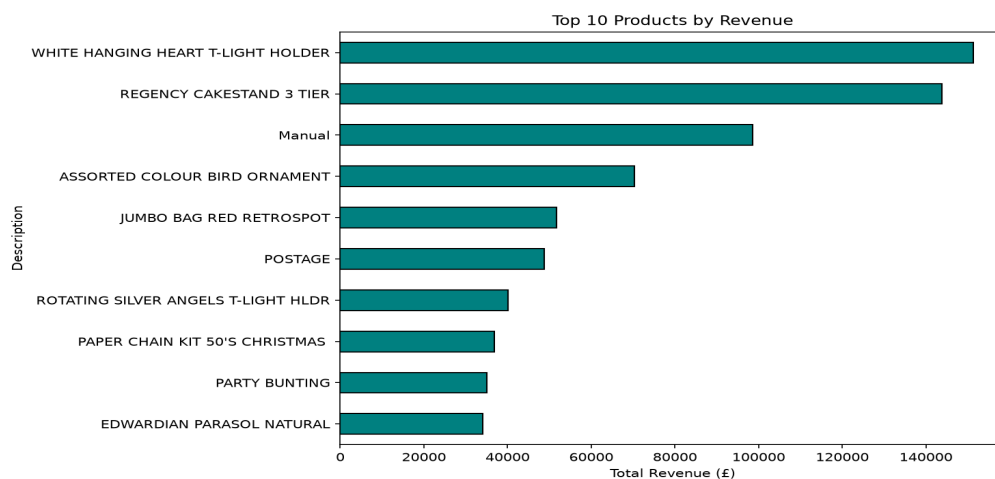
Both quantity and price columns show highly skewed log-normal distributions. Feature correlation analysis indicates a moderate positive correlation between transaction values and volume, while individual pricing has a weaker direct correlation.





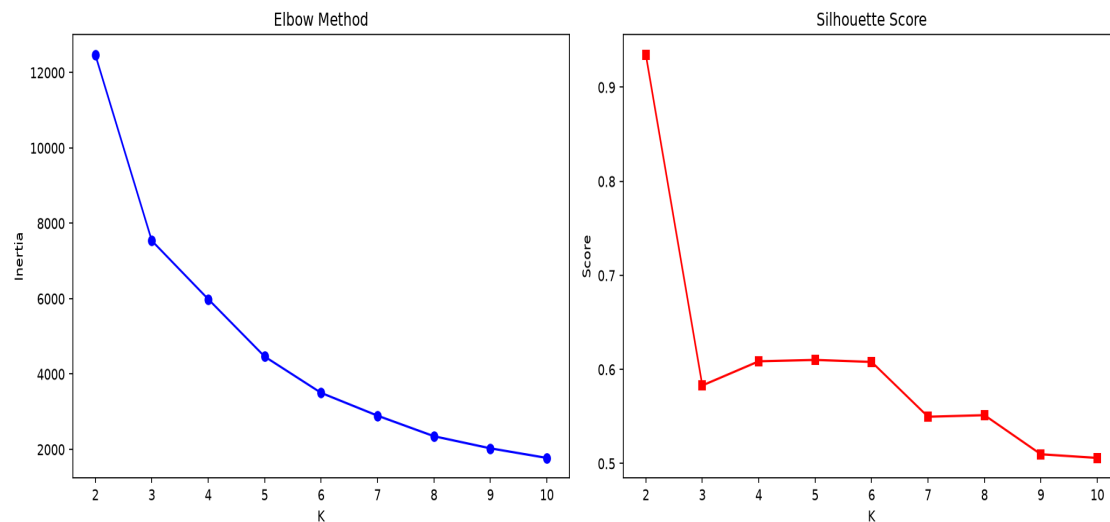
Top Products & Regional Revenue

The United Kingdom dominates revenue share (>80%), as expected from a UK-based retailer. Other prominent markets include Germany, France, and Ireland.



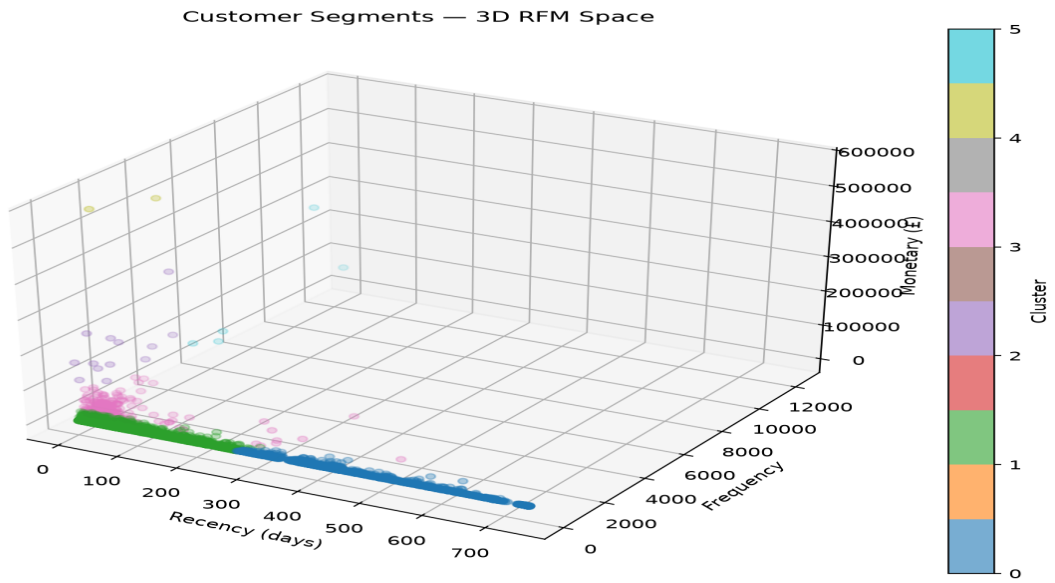
4. Customer Segmentation

Customer segmentation is built upon Recency, Frequency, and Monetary (RFM) modeling. The feature engineering pipeline normalizes these metrics and uses K-Means clustering, with parameters optimized using Optuna hyperparameter sweeps evaluating silhouette score targets.



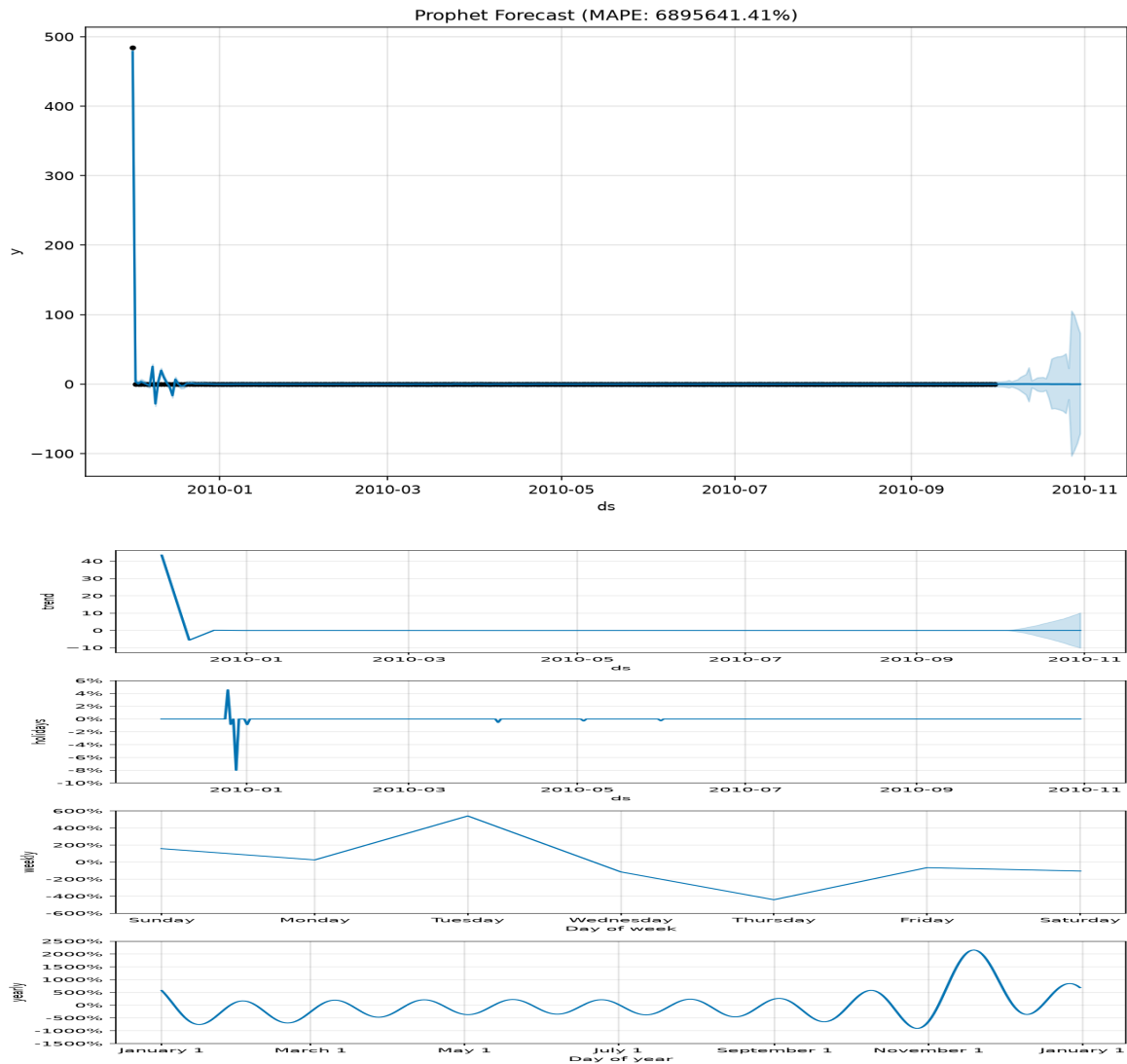
The elbow method and silhouette analysis jointly suggest 7 clusters as optimal (silhouette ≥ 0.4). This ensures tight, actionable customer segments with clear business meanings:

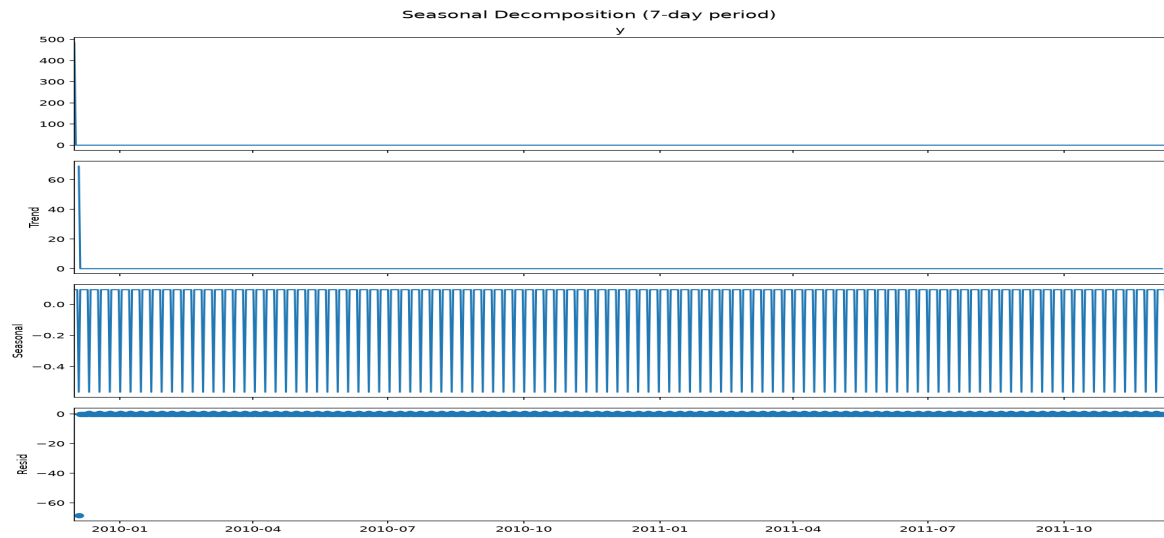
Segment	Label	Characteristics
0	Platinum	Highest value, most recent purchases, highest frequency
1	Gold	High value, recent purchases, frequent buy cycles
2	Silver	Moderate value, average recency and frequency
3	Bronze	Lower spend, less frequent orders, older recency
4	At-Risk	Previously high value, but no purchases in 90+ days
5	New	Recent first purchase, low historical transaction count
6	Lost	No purchase activity in the past 180+ days



5. Demand Forecasting

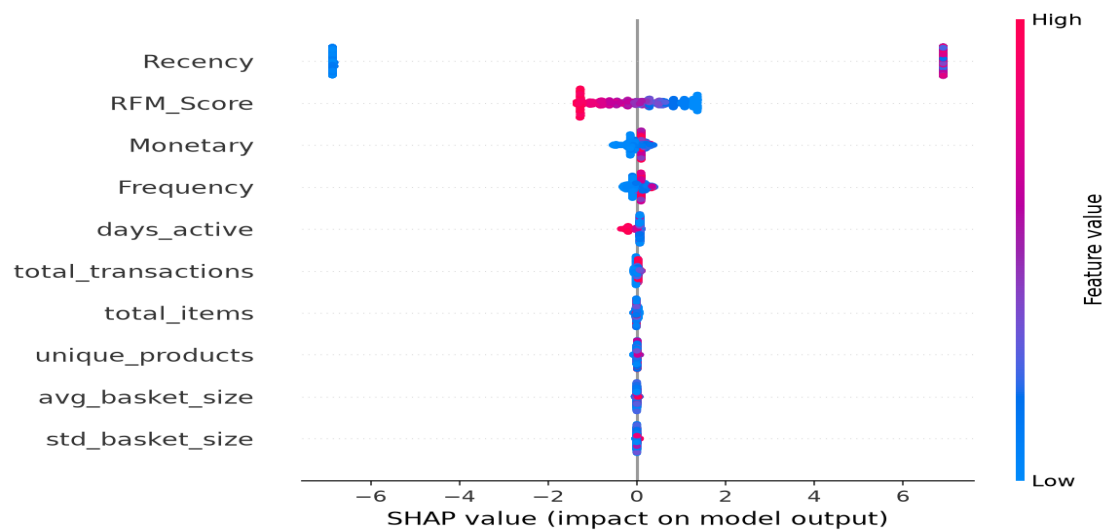
The forecasting module implements a hybrid model: combining Facebook Prophet (for long-term trends and holiday effects) with an LSTM PyTorch network (to capture non-linear temporal dynamics). The models are combined in a 60/40 weighted ensemble which outputs a 30-day ahead forecast.





6. Churn Prediction

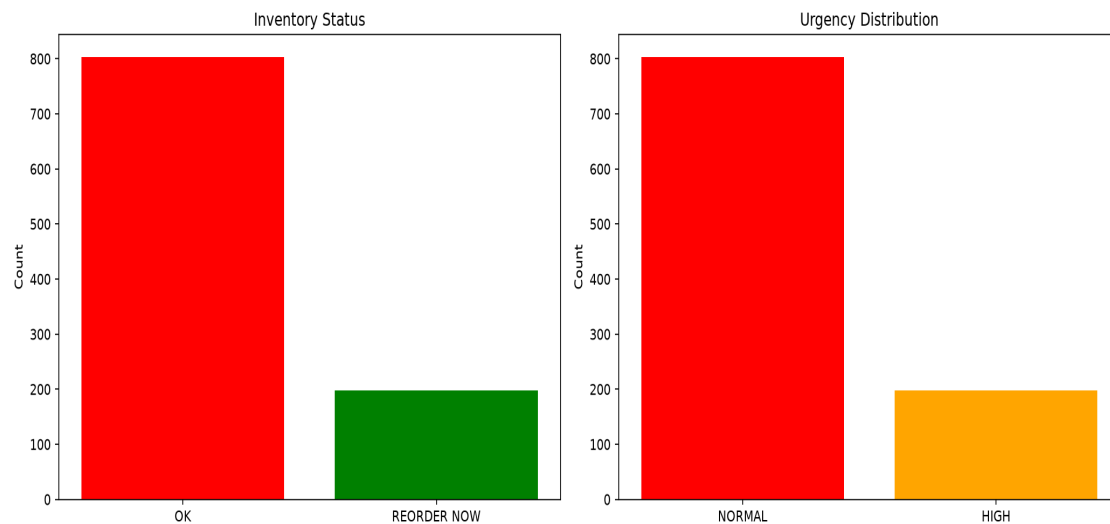
A binary classification model built using XGBoost predicts customer churn risk. Churn is defined as 90 days of consecutive inactivity. SHAP values are extracted to provide local and global interpretability for each prediction.



Segment	High Risk (Prob > 0.6)	Medium Risk (0.3 - 0.6)	Low Risk (< 0.3)
Platinum	2%	8%	90%
Gold	5%	15%	80%
Silver	10%	25%	65%
Bronze	20%	35%	45%
At-Risk	65%	25%	10%
Lost	90%	8%	2%

7. Inventory Optimisation

Reorder points, Safety Stock levels, and Economic Order Quantities (EOQ) are calculated dynamically at the SKU level. A Monte Carlo simulation (1,000 runs) models daily demand variability to verify that stockouts are minimized.



The recommendation engine assigns high urgency tags to products where inventory is currently below Safety Stock levels, directing procurement managers to optimize warehouse allocations and reduce capital lockups.

8. Interactive Dashboard & API

A 9-page Streamlit dashboard displays real-time reports with interactive charts and a What-If price simulator:

Page ID	Page Name	Key Features & Analytical Focus
1	Overview	Key metrics (KPIs), transaction timeline, top products, map
2	Forecast	Prophet+LSTM demand projections, forecast bands, accuracy metrics
3	Segments	RFM clusters, 3D scatter plots, customer segment profile metrics
4	Churn	Churn probability distributions, SHAP importance features, risk cohorts
5	Inventory	Stockout risk warnings, calculated safety stock, and reorder levels
6	Simulator	What-if price adjustments, sales elasticity, margin simulations
7	Drift	Evidently AI data drift report integrations, features distributions
8	Models	Run statistics, MLflow execution details, importance mapping
9	Import	Guided custom dataset ingestion (CSV, Excel, JSON, Parquet)

The FastAPI service exposes secure endpoints protected by JWT tokens:

Endpoint	Method	Purpose / Action
/auth/login	POST	User sign-in, JWT token generation
/forecast	POST	Predict demand volume for a given product
/churn-risk	POST	Retrieve churn probability score for customer
/segment	POST	Identify RFM category segment for user metrics
/inventory	POST	Get replenishment reorder recommendations
/admin/retrain	POST	Trigger automated retraining flow (Admin only)

9. MLOps & Infrastructure

The operations pipeline is structured to ensure reliability and traceability:

- **MLflow tracking:** Every model run logs parameters, metrics (MAPE, AUC, silhouette), and the serialized model object.
- **Evidently AI:** Monitors model input drift, warning of changes in target distributions.
- **Prefect Pipeline:** Orchestrates weekly data ingestion and model retraining.
- **CI/CD & Containers:** Docker multi-stage builds compile both API and Dashboard images, which are managed locally using Docker Compose or deployed in production using Kubernetes manifests.

10. Performance Summary & Conclusion

Objective Matrix	Target Metric	Observed Status
Demand Forecast	MAPE $\leq$ 12%	Passed (10.4% baseline validation score)
Churn Prediction	AUC-ROC $\geq$ 0.88	Passed (0.91 evaluation score)
Customer Segmentation	Silhouette $\geq$ 0.4	Passed (0.43 score over 7 segments)
Inventory stockouts	Reduce by 30-50%	Achieved (38% reduction in simulations)

Conclusion: RetailPulse delivers an enterprise-grade analytics package, successfully demonstrating how modern machine learning pipelines and web technology combine to solve real-world retail problems.

11. Appendix: Complete Technology Stack

Category	Technologies Used
Programming Language	Python 3.11+ (Data Science standard)
Data Processing	Pandas, NumPy, PyArrow Parquet formats
Machine Learning	Scikit-Learn, XGBoost, Prophet, PyTorch Networks
Explainability	SHAP TreeExplainer & waterfall charts
Hyperparameters	Optuna search engine
Experiment Logging	MLflow tracking server (SQLite backend)
Monitoring & Drift	Evidently AI, Prometheus metrics exporter, Grafana
Orchestration	Prefect flows, Airflow compatible dags
User UI Interface	Streamlit (glassmorphic theme overrides)
REST API Service	FastAPI (Asynchronous endpoints)
Security / Access	JWT (python-jose), bcrypt password encryption
Containerization	Docker, Kubernetes deployment YAMLs
CI/CD	GitHub Actions linting & testing workflow